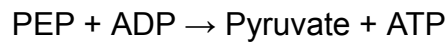


A newborn girl is diagnosed with a rare genetic disorder that causes premature lysis of red blood cells. Through genome sequencing, physicians identified a specific mutation (R510Q) in both copies of her pyruvate kinase gene, which encodes the enzyme that catalyzes the final step of glycolysis. Pyruvate kinase produces pyruvate and ATP from phosphoenolpyruvate (PEP) and ADP.



Reaction 1

Because red blood cells lack mitochondria, pyruvate in erythrocytes does not enter the citric acid cycle and the electron transport chain. Therefore, red blood cells rely almost exclusively on glycolysis to produce sufficient ATP for cellular functions. In patients with pyruvate kinase deficiency, a lack of ATP is believed to compromise the integrity of the red blood cell membrane, leading to premature lysis in the spleen. Although these observations are well established, the exact mechanism by which low ATP alters the cell membrane is unclear.

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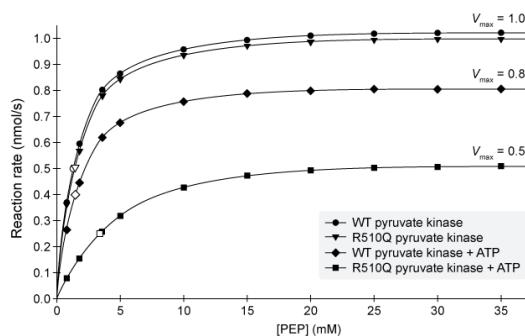


Figure 1 Pyruvate kinase activity with and without ATP as a function of [PEP]. Open markers are shown at $\frac{1}{2}V_{\max}$ for each plot.

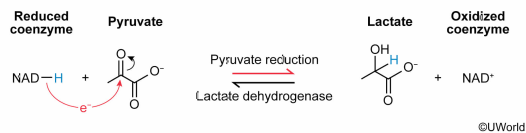
Based on information in the passage, pyruvate in red blood cells can be further metabolized only by:

- ☐ A. phosphorylation of ADP to form ATP. [21%]
- ☐ B. decarboxylation to form acetyl-CoA. [9%]
- ☒ C. reduction to form lactate and NAD^+ . [64%]
- ☐ D. carboxylation to form oxaloacetate. [3%]

Correct

64% answered correctly

Explanation:



Glycolysis breaks down glucose into pyruvate in a series of reactions that generate ATP and reduce NAD⁺ to NADH. **NAD⁺ must be regenerated** for glycolysis to continue, and this process can occur either in the mitochondria (electron transport chain) or in the cytosol (fermentation).

According to the passage, red blood cells lack mitochondria, and therefore they can regenerate NAD⁺ only by fermentation. Fermentation occurs when the enzyme **lactate dehydrogenase** catalyzes the **reduction of pyruvate to lactate**. The required electrons are provided by the oxidation of NADH to form **NAD⁺**.

(Choice A) Phosphorylation of ADP to form ATP occurs when pyruvate is synthesized by pyruvate kinase, not when pyruvate is subsequently metabolized.

(Choice B) Pyruvate is decarboxylated in mitochondria to form acetyl-CoA. Because red blood cells do not have mitochondria, they cannot process pyruvate by this pathway.

(Choice D) Carboxylation of pyruvate to form oxaloacetate is the first step in **gluconeogenesis**. However, gluconeogenesis occurs in the liver, not in red blood cells.

Educational objective:

In the absence of mitochondria, the NAD⁺ needed for glycolysis is regenerated when pyruvate is reduced to lactate. This process is catalyzed by the enzyme lactate dehydrogenase.

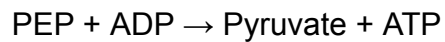
Time spent: 0 sec

Subject:
Biochemistry

QID: 401579

System:
Enzymes

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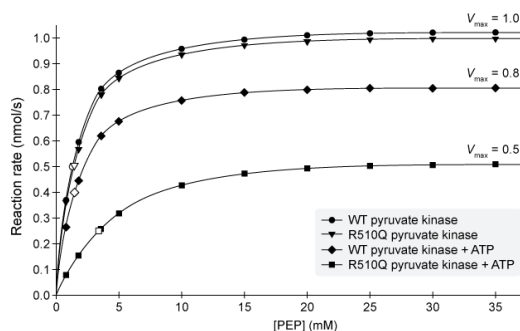
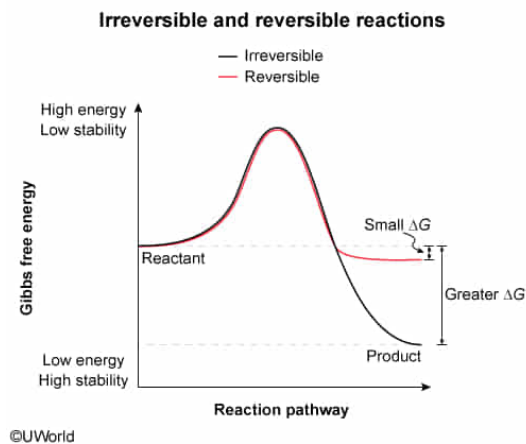


Figure 1 Pyruvate kinase activity with and without ATP as a function of [PEP]. Open markers are shown at $\frac{1}{2}V_{\max}$ for each plot.

Given that pyruvate synthesis is generally unidirectional, which of the following statements regarding the thermodynamics of Reaction 1 is most accurate?

- ☐ A. The enthalpy change is positive. [6%]
- ☐ B. The enthalpy change is negative. [9%]
- ☐ C. The net free energy change is positive. [11%]
- ☒ D. The net free energy change is negative. [72%]

Explanation:

The direction in which a chemical reaction tends to proceed depends on the net free energy change between reactants and products, referred to as the **Gibbs free energy change ΔG** . The natural tendency of a reaction is to minimize the Gibbs free energy, so compounds in a lower energy state (more stable) are more likely to form under a given set of conditions than those in high energy states.

Favorable reactions have **negative** net free energy changes ($-\Delta G$) because the products have decreased free energy relative to the reactants, and they proceed **spontaneously**. In contrast, unfavorable reactions have positive free energy changes ($+\Delta G$) and form less stable products; these reactions are nonspontaneous and require an outside source of energy to proceed.

The synthesis of pyruvate from phosphoenolpyruvate is classified as a unidirectional reaction in the question. Reactions with a **large free energy change** between the products and reactants are **typically unidirectional** because the reverse reaction is so unfavorable that it essentially never occurs. Therefore, pyruvate synthesis is most likely highly thermodynamically favorable and must have a **negative free energy change**.

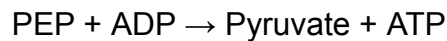
(Choices A and B) The enthalpy change ΔH of a reaction reflects the heat released or consumed by the reaction, not its ability to proceed. Although enthalpy is related to the free energy of a reaction by the equation $\Delta G = \Delta H - T\Delta S$ (where ΔG is the change in free energy, T is the temperature in kelvin, and ΔS is the change in **entropy**), it is possible for a reaction to have a negative free energy regardless of its enthalpy.

(Choice C) Positive free energy corresponds to an *unfavorable* reaction that tends to proceed in the reverse direction.

Educational objective:

Whether a reaction proceeds is dependent on the Gibbs free energy change ΔG of that reaction. A negative free energy change ($-\Delta G$) is associated with a favorable reaction that proceeds spontaneously whereas a positive free energy change requires outside energy to drive the reaction to completion. Reactions with large free energy changes are essentially irreversible, and therefore are considered unidirectional.

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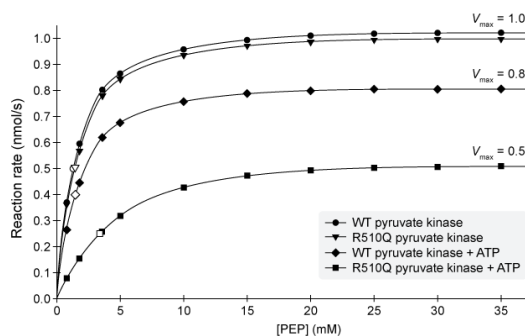


Figure 1 Pyruvate kinase activity with and without ATP as a function of [PEP]. Open markers are shown at $\frac{1}{2}V_{\max}$ for each plot.

Based on Figure 1, which of the following statements is true regarding R510Q pyruvate kinase as compared to WT pyruvate kinase?

- ☐ A. WT and R510Q pyruvate kinase have approximately the same K_M in all conditions but with varying k_{cat} . [5%]
- ☐ B. R510Q pyruvate kinase has a lower K_M in the presence of ATP than in the absence of ATP. [8%]
- ☒ C. WT and R510Q pyruvate kinase have about the same k_{cat} and K_M in the absence, but

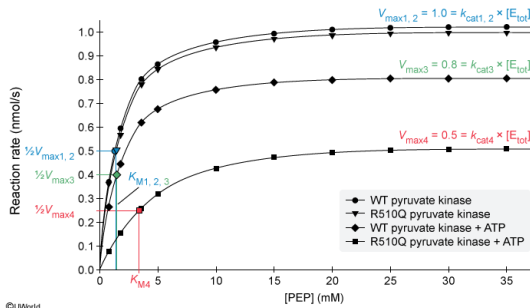
not the presence, of ATP. [83%]

- ☐ D. WT pyruvate kinase has a greater decrease in k_{cat} than does R510Q pyruvate kinase when ATP is added. [1%]

Correct

83% answered correctly

Explanation:



Enzyme kinetics are frequently described using the [Michaelis-Menten equation](#), which calculates the reaction velocity at varying substrate concentrations. On a graph of reaction velocity V_0 versus substrate concentration $[S]$, the **maximum velocity** V_{max} is found at the plateau of the curve. V_{max} represents the reaction rate when an enzyme's active sites are completely saturated with substrate, and it is directly proportional to the total enzyme concentration and to the turnover number k_{cat} . On a graph, V_{max} corresponds to the maximum height of the curve.

The **Michaelis constant** K_M is the **substrate concentration** at which $\frac{1}{2}V_{\text{max}}$ is achieved.

Therefore, K_M values can be found by identifying the substrate concentration at which the curve reaches [half its plateau height](#). On a graph, K_M can be identified by drawing a line from the point on the curve where $\frac{1}{2}V_{\text{max}}$ is achieved down to the x-axis. The x-axis value (substrate concentration) at this point is equal to the K_M value.

Figure 1 shows reaction velocity versus $[S]$ plots for WT and R510Q pyruvate kinase in the presence and absence of ATP. **In the absence of ATP**, WT and R510Q pyruvate kinase **behave almost identically**. Both enzymes have similar V_{max} and K_M values because their curves plateau at approximately the same height and reach half the maximum height at approximately the same substrate concentration. Because the reactions are run at equal enzyme concentrations, the similar V_{max} values also indicate similar k_{cat} values. However, in the **presence of ATP**, WT pyruvate kinase has a **higher** V_{max} (and k_{cat}) and a **lower** K_M than R510Q pyruvate kinase (ie, R510Q K_M is right-shifted).

(Choices A and B) In the presence of ATP, the K_M value of R510Q pyruvate kinase *increases* rather than decreasing or remaining the same.

(Choice D) When ATP is added, the V_{max} (and therefore the k_{cat}) of R510Q pyruvate kinase is much lower than that of WT pyruvate kinase.

Educational objective:

Enzyme activity is expressed mathematically by the Michaelis-Menten equation, $V_0 = \frac{V_{\max}[S]}{K_M + [S]}$. $V_{\max}$ is directly proportional to k_{cat} and to enzyme concentration, and it is indicated by the height of the plateau on a Michaelis-Menten graph. The Michaelis constant K_M is the substrate concentration at which $\frac{1}{2}V_{\max}$ is achieved.

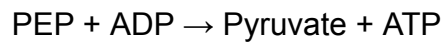
Time spent:0 sec

Subject:
Biochemistry

QID:401581

System:
Enzymes

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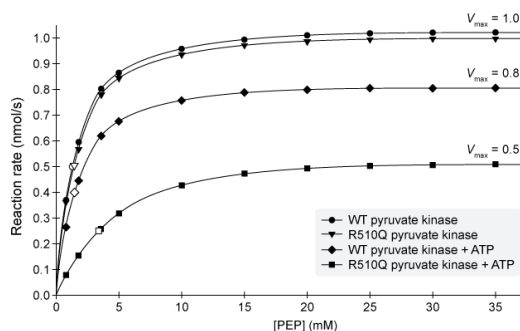


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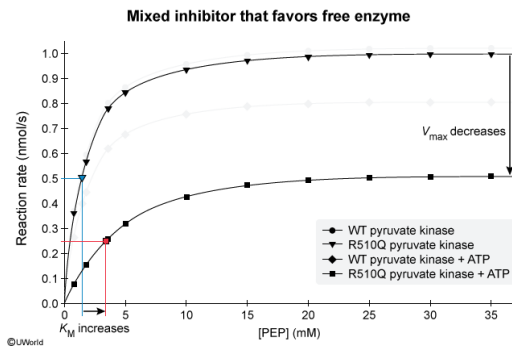
The information in the passage and the data in Figure 1 support which of the following conclusions about ATP as an inhibitor of the R510Q enzyme?

- ☐ A. ATP can bind only free pyruvate kinase. [3%]
- ☐ B. ATP binds free pyruvate kinase and the enzyme-substrate complex with equal affinity. [29%]
- ☐ C. ATP can bind only the enzyme-substrate complex. [9%]
- ☒ D. ATP binds both free pyruvate kinase and the enzyme-substrate complex, but it favors free enzymes. [57%]

Correct

57% answered correctly

Explanation:



Inhibitors have a **spectrum of effects** on enzymes. **Competitive inhibitors** bind only **free enzymes** and prohibit substrate from binding. This leads to an **increased K_M** (more substrate required to reach $\frac{1}{2}V_{max}$) but no change in V_{max} . **Uncompetitive inhibitors** bind only the **enzyme-substrate (ES) complex**. The formation of an ES-inhibitor complex effectively removes ES complexes from the system, and therefore shifts the equilibrium toward additional ES complex formation. This action causes a **decrease in K_M** and a proportional decrease in V_{max} .

Mixed inhibitors bind both free enzymes and the ES complex, and therefore they have characteristics of both competitive and uncompetitive inhibitors (ie, they are a mix of competitive and uncompetitive). All mixed inhibitors cause a decreased V_{max} but have distinct effects on K_M , depending on their relative affinities for free enzymes and ES complexes:

- Inhibitors with a **greater affinity for free enzymes** are more similar to competitive inhibitors and cause an *increase* in K_M .
- Inhibitors with a **greater affinity for ES complexes** are more like uncompetitive inhibitors and cause a *decrease* in K_M .
- Mixed inhibitors that have equal affinity for free enzymes and the ES complex represent a special case called **noncompetitive inhibition**. Noncompetitive inhibitors have *no effect* on K_M .

Figure 1 shows that ATP *decreases* the V_{max} of R510Q pyruvate kinase and *increases* K_M .

Therefore, ATP is acting as a **mixed inhibitor** with a higher affinity for free enzymes than for the ES complex.

(Choice A) If ATP could bind only free pyruvate kinase (competitive inhibition), V_{max} would remain unchanged.

(Choice B) ATP binds both free **WT enzymes** and the ES complex with equal affinity (noncompetitive inhibition), but this is not true for R510Q enzymes.

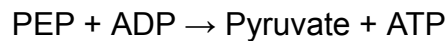
(Choice C) If ATP could bind only the ES complex (uncompetitive inhibition), V_{max} and K_M would both decrease by *exactly* the same factor.

Educational objective:

Competitive inhibitors bind only free enzymes whereas uncompetitive inhibitors bind only the enzyme-substrate (ES) complex. Mixed inhibitors can bind both free enzymes and the ES complex, and the special case where the inhibitor binds both with equal affinity is called noncompetitive inhibition. Preferentially binding free enzymes causes an increase in K_M ; preferentially binding the ES complex causes a decrease in K_M ; and when both are bound equally, K_M is unaffected.

Time spent:0 sec**Subject:**
Biochemistry**QID:**401582**System:**
Enzymes

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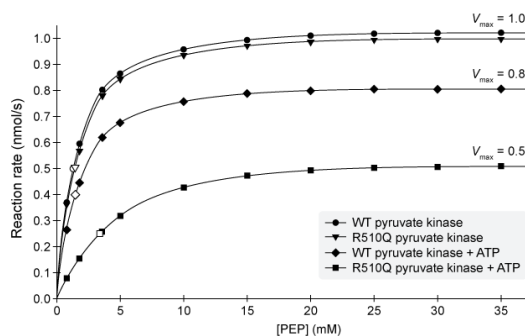


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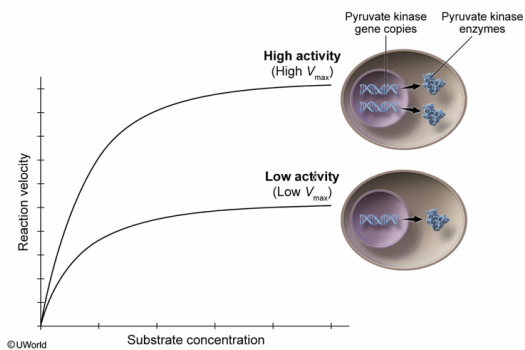
Scientists predict that gene copy number is proportional to protein expression for pyruvate kinase. If this hypothesis is correct, which kinetic parameter would be expected to double when the gene copy number doubles?

- ☐ A. Catalytic efficiency [16%]
- ☒ B. Maximum velocity [46%]
- ☐ C. Catalytic turnover [31%]
- ☐ D. Equilibrium constant [5%]

Correct

46% answered correctly

Explanation:



The rate of an enzyme-catalyzed reaction plateaus at high substrate concentrations ($[S] \gg K_M$) because the active sites of enzymes are completely saturated with substrate; adding more substrate fails to increase the rate of catalysis. Adding more enzyme, on the other hand, provides additional active sites and allows the reaction to proceed more quickly as more substrate can be bound.

The **maximum velocity** V_{max} of the reaction is the product of the number of active sites (**total enzyme concentration** $[E_{tot}]$) and the rate at which each site converts substrate to product (**turnover number** k_{cat}). Mathematically, this value is described as $V_{max} = k_{cat} \times [E_{tot}]$. Therefore, an increase in $[E_{tot}]$ will increase V_{max} .

The question states that the gene copy number is predicted to be proportional to protein expression. If this prediction is correct, doubling the gene copy number should result in double the total enzyme purified from a given number of cells. As a result, doubling pyruvate kinase's gene copy number should also **double the** V_{max} of the pyruvate kinase reaction.

(Choice A) Catalytic efficiency is the ratio of k_{cat} to the Michaelis constant K_M . Both are intrinsic properties of the enzyme and are not altered by a change in enzyme concentration.

(Choice C) Catalytic turnover k_{cat} is an intrinsic property of the enzyme and is not altered by a change in enzyme concentration.

(Choice D) $[E_{tot}]$ and therefore the gene copy number has no effect on the **equilibrium constant** K_{eq} of a reaction. Enzymes increase the rate at which equilibrium is achieved but do not affect the equilibrium itself.

Educational objective:

At high substrate concentrations ($[S] \gg K_M$), the active sites of enzymes are completely saturated with substrate, and adding more substrate does not increase the reaction rate. Adding more enzyme increases the number of active sites, and therefore increases the reaction velocity. The maximum velocity at any enzyme concentration is calculated as $V_{max} = k_{cat} \times [E_{tot}]$.

Time spent: 0 sec

Subject:

Biochemistry

QID: 401583

System:

Exam

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